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ELECTRICAL CONNECTOR

Abstract

An electrical connector includes a base and a plurality of contact modules. The base has a plurality of tunnels. The contact modules respectively embedded in the tunnels, wherein each of the contact modules includes a contact and a spring. The spring is located in one of the tunnels and supports a contact end of the contact so that the contact end is suspended. A back end of the spring has a fixed portion having a first barb, and the first barb hooks into a stop recess of a first tube wall of one of the tunnels.

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Background/Summary

BACKGROUND

Field of Invention

[0001] The present disclosure relates to an electrical connector.

Description of Related Art

[0002] Current high-power transmission connectors often use high-conductivity copper, known as red copper, for making terminals that improve electrical conductivity.

[0003] Yet, red copper's softness and inadequate fatigue strength can cause it to wear out quickly in plug-in uses, shortening its lifespan.

[0004] A common solution is using a spring clip under the soft terminal to help support and distribute the load.

[0005] To assemble these, a spring clip is first fitted into a plastic insulating base using fixture(s). Heat is then applied to form protrusions in the base. The terminal is then inserted, held in place by these protrusions, and a groove on the terminal locks with the spring for a secure fit. This method, though, is complicated and often unstable. Moreover, by the said process, the terminal's stability depends on the spring's position within the base, either deformation of terminal or spring may result in the failure of or misalignment occurs, it poses a risk of terminal detachment, adversely affecting product quality.

[0006] Additionally, the terminal's stability relies on the spring's exact placement inside the base. If the terminal and spring are not aligned correctly or if either is deformed, the terminal can come loose, affecting the quality of the product.

SUMMARY

[0007] According to some embodiments of the present disclosure, an electrical connector includes a base and a plurality of contact modules. The base has a plurality of tunnels. Each of the contact modules includes a contact and a spring. A back end of the spring has a fixed portion having a first barb. The first barb of the spring is hooked into the stop recess of the tunnel of the base for secure, and the spring may have an enclosed ring for allowing the contact to pass through. Moreover, another barb may be formed on the surface of the enclosed ring. By the said design, the product assembly process required thereby can be simplified, making it easy to assemble with high production efficiency, and ensuring a stable product structure with a long service life. The aforementioned configuration has the following advantages. The contact and the spring form the easy-to-assemble contact module, the first barb of the spring inserted into the terminal hole of the base and coupled with the stop recess realizes the assembly and the limiting of the entire contact module, and thus complex manufacturing processes can be decreased to simplify the assembly process of the product. In addition, the spring can not only be used as a limiting fixture between the contact and the base to ensure that the position of the contact in the base is stable, but also can provide elastic support for the contact, so that the contact can indirectly obtain better elasticity, thereby greatly increasing the plugging and unplugging lifespan of the product.

[0008] In the aforementioned embodiments of the present disclosure, by combining the terminal and spring clip into an easily assembled terminal module. The module is secured in place within the base by aligning the spring clip's first barb with the terminal hole and a stop groove. This design simplifies the assembly process, reducing complex steps and making it more straightforward. Additionally, the spring serves a dual function that it acts as a positioning and stabilizing component between the terminal and the base, ensuring the terminal remains securely in place, and it provides elastic support to the terminal. This arrangement indirectly enhances the terminal's elasticity, significantly extending the product's plug-in lifespan.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the

standard practice in the industry, various features are drawn accurately according to the real scale, so the details such as the proportion and relative relationship of each element should be regarded as a part of the content of the present disclosure.

[0010] FIG. 1 is a rear view of an electrical connector according to one embodiment of the present disclosure.

[0011] FIG. 2 is a cross-sectional view of the electrical connector of FIG. 1.

[0012] FIG. 3 is a perspective view of a contact module of the electrical connector of FIG. 2.

[0013] FIG. 4 is a side view of the contact module of FIG. 3.

[0014] FIG. 5 is a schematic view of a contact and a spring of the contact module of FIG. 4 during assembling.

[0015] FIG. 6 is a partially enlarged view of the contact module of FIG. 4.

[0016] FIG. 7 is a cross-sectional view of the electrical connector of FIG. 2 when being electrically coupled with a mating connector.

DETAILED DESCRIPTION

[0017] The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

[0018] Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

[0019] It is noted that various features are drawn accurately according to the real scale, so the details such as the proportion and relative relationship of each element should be regarded as a part of the content of the present disclosure.

[0020] FIG. 1 is a rear view of an electrical connector **100** according to one embodiment of the present disclosure. FIG. 2 is a cross-sectional view of the electrical connector **100** of FIG. 1. As shown in FIG. 1 and FIG. 2, the electrical connector **100** may be a high current and high voltage connector, such as an electrical connector capable of transmitting 40 A and 400V power. In this embodiment, the electrical connector **100** is a female connector configured to couple with a mating connector, such as a male connector.

[0021] The electrical connector **100** includes a base **110** and plural contact modules **130**.

[0022] The base **110** is one-piece-formed of electrically insulated material. For example, the base **110** is made of plastic by injection molding. The base **110** has three tunnels **111** arranged in a pyramid shape. Each of the tunnels **111** has a lower first tube wall **112** and an upper second tube wall **114**. The first tube wall **112** has a stop recess **113**, and the second tube wall **114** has a stop recess **115**. In this embodiment, the stop recess **113** is a through hole, and the stop recess **115** is a blind hole. In this embodiment, the stop recess **115** can be concentric to the stop recess **113**. In another embodiment, either or both of the said recesses may also be formed at the lateral surfaces instead.

[0023] In addition, as depicted in FIG. 2, the base **110** has a step structure **116** in the tunnel **111**, and back side opening of the tunnel **111** of the base **110** has two wing-shaped grooves **117** (see FIG. 1) formed on both sides of the opening. Each of the contact modules **130** is respectively embedded in the tunnels **111** of the base **110** respectively.

[0024] FIG. 3 is a perspective view of the contact module 130 of the electrical connector 100 of FIG. 2. FIG. 4 is a side view of the contact module 130 of FIG. 3. As shown in FIG. 3 and FIG. 4, the contact module 130 includes a contact 131 and a spring 141. Each of the contact 131 may be one-piece-formed by a metal plate, and be formed of such as red copper.

[0025] The contact 131 is fork shaped and has an ear portion 132 and a positioning recess 133. The ear portions 132 protrude from the lateral surface/sidewall of the contact 131 respectively. The positioning recess 133 forms on the bottom surface of a back end thereof as depicted by FIG. 4.

[0026] The contact end 135 of the contact 131 has a curved structure. In the following description, the front end and the back end of the contact module 130 are respectively referred to as the left side and the right side of the contact module 130 shown in FIG. 4. As depicted in FIG. 3, the back end of the contact 131 has a through hole 136 for electrically connecting a wire (e.g., by soldering).

[0027] The spring 141 is one-piece-formed by a metal material with high strength, for example, SUS301-H stainless steel. The spring 141 is integrally formed as a single piece by a punching process or a stamping process.

[0028] The spring 141 from the front end to the back end, in sequence, has a contact portion 1411, an extending portion 1412, a turning portion 1413, a fixed portion 1414, and a tail portion 1415.

[0029] The front end of the spring 141 includes the contact portion 1411, and the spring 141 includes the extending portion 1412 behind the contact portion 1411. In this embodiment, the contact portion 1411 of the spring 141 has a fork shape with a slot to correspond to the shape of the contact end 135 of the contact 131. The formation of the slot is optional, and the contact portion 1411 of the spring 141 may have a continuous surface without the slot in another embodiment.

[0030] The contact portion 1411 of the spring 141 has a curved structure.

[0031] Each of two sides of the extending portion 1412 of the spring 141 has a side wing 148, and the back of the side wing 148 has a guiding surface 149. The spring 141 includes the turning portion 1413 behind the extending portion 1412. The back of the turning portion 1413 connects the fixed portion 1414. Therefore, the spring 141 includes the contact portion 1411 included by the front end of the spring 141, the extending portion 1412, the turning portion 1413, and the fixed portion 1414 in sequence.

[0032] The back end of the spring 141 has the fixed portion 1414 having a first barb 1414A. The fixed portion 1414 is in a form of an enclosed ring 144. The surface of the fixed portion 1414 in the form of the enclosed ring 144 facing away from the first barb 1414A has a second barb 1414B. In this embodiment, the second barb 1414B is substantially aligned with the first barb 1414A along a vertical direction. Moreover, the fixed portion 1414 of the spring 141 is lower than the extending portion 1412 of the spring 141.

[0033] In addition, the back of the fixed portion 1414 connects the tail portion 1415 of the spring 141. In this embodiment, the tail portion 1415 of the spring 141 extends upward, such as an inclined sidewall.

[0034] It is to be noted that the connection relationships, the materials, and the advantages of the elements described above will not be repeated in the following description. In the following description, other connection relationships between different elements and the assembling steps of the electrical connector 100 will be explained.

[0035] FIG. 5 is a schematic view of the contact 131 and the spring 141 of the contact module 130 of FIG. 4 during assembling. FIG. 6 is a partially enlarged view of the contact module of FIG. 4. As shown in FIG. 5 and FIG. 6, the enclosed ring 144 of the spring 141 forms a channel to enable the contact 131 to pass through. Thereafter, the contact 131 can be pushed in a direction D1 to slide along the spring 141 until the tail portion 1415 of the spring 141 extends upward into the positioning recess 133 of the surface of the contact 131. The side wing 148 and the enclosed ring 144 of the spring 141 prevent the contact 131 from shifting in directions D2 and D3. In this state, the ear portion 132 of the contact 131 abuts against the sidewall of the fixed portion 1414 in the form of the enclosed ring 144, and the curved structure of the contact portion 1411 of the spring

141 abuts against the curved structure of the contact end **135** of the contact **131**. The ear portion **132** of the contact **131** abutting against the enclosed ring **144** of the spring **141** and the tail portion **1415** of the spring **141** extending into the positioning recess **133** of the contact **131** prevent the contact **131** from shifting in the direction **D1**. In this embodiment, the bottom curved surface of the contact end **135** of the contact **131** is disposed along the top curved surface of the contact portion **1411** of the spring **141**. The spring **141** supports the contact end **135** of the contact **131** so that the contact end **135** is suspended.

[0036] Referring back to FIG. 1 and FIG. 2, after the contact **131** is assembled to the spring **141** to form the contact module **130**, the contact module **130** is inserted into the opening of the tunnel **111** of the base **110**, and the groove **117** of the base **110** corresponds to the shape of the ear portion **132** of the contact **131**, such that the ear portion **132** of the contact **131** can be coupled with the groove **117** of the base **110**. The ear portion **132** and the groove **117** can facilitate the positioning of the contact module **130** within the base **110**. Thereafter, the spring **141** is located in the tunnel **111** of the base **110**, the first barb **1414A** hooks into the stop recess **113** of the first tube wall **112** of the tunnel **111**, and the second barb **1414B** hooks into the stop recess **115** of the second tube wall **114** of the tunnel **111**. In other words, the position of the stop recess **113** corresponds to the position of the first barb **1414A**, and the position of the stop recess **115** corresponds to the position of the second barb **1414B**. As a result, the position of the contact module **130** is limited by the first barb **1414A**, the second barb **1414B**, and the two stop recesses **113** and **115**.

[0037] In this state, the contact portion **1411** of the spring **141** suspended from the bottom wall (e.g., the first tube wall **112**) of the tunnel **111**, the extending portion **1412** of the spring **141** is at least flat against the bottom wall of the tunnel **111**, the turning portion **1413** of the spring **141** corresponds to the step structure **116** of the base **110**, and the fixed portion **1414** of the spring **141** is partially flat against the first tube wall **112**. The turning portion **1413** and the step structure **116** can facilitate the positioning of the contact module **130** within the base **110** in a direction **D4**. In this embodiment, the shape of the turning portion **1413** corresponds to the shape of the step structure **116**, such that the turning portion **1413** of the spring **141** can be disposed along the step structure **116** of the base **110**. In addition, the side wing **148** of the extending portion **1412** of the spring **141** extends in a direction away from the first tube wall **112** (i.e., in an upward direction).

[0038] FIG. 7 is a cross-sectional view of the electrical connector **100** of FIG. 2 when being electrically coupled with a mating connector **200**. The mating connector **200** may be a male connector configured to couple with the electrical connector **100**. The mating connector **200** has a base **210** a contact **230**. When the mating connector **200** and the electrical connector **100** are mated, the contact **230** is in electrical contact with the contact end **135** of the contact **131**, in which the contact **131** is flexible and is placed under tension by the spring **141** so that the shaped region of the contact **131** rides over the leading end of the contact **230** and press against the bottom surface of the contact **230**. Although FIG. 7 merely shows that the contact **230** abuts against the contact **131**, the contact **131** may move slightly downwards due to the flexibility of the underlying spring **141**.

[0039] Specifically, because the contact module **130** of the electrical connector **100** has the spring **141** to support the contact end **135** of the contact **131**, the contact end **135** of the contact **131** can be suspended to ensure that the contact end **135** of the contact **131** is in electrical contact with the contact **230** of the mating connector **200**. The spring **141** serves as a limiting fixture for the contact **131**, and provides the contact **131** elastic support, thereby extending plug and unplug lifespan. Moreover, since the fixed portion **1414** of the spring **141** has the first and second barbs **1414A** and **1414B**, and the tunnel **111** of the base **110** has the stop recesses **113** and **115**, the spring **141** and the contact **131** on the spring **141** can be positioned in the tunnel **111** of the base **110** by the first and second barbs **1414A** and **1414B** and the stop recesses **113** and **115** of the tunnel **111**. The electrical connector **100** is convenient to assemble and its structure is stable and reliable, and can suppress electrical arcing between separated contacts during connection and disconnection without adding

an additional connection and switching device.

[0040] In detail, the steps of the installation and the use of the present disclosure are: [0041] a. the contact **131** is assembled to the spring **141** to form the contact module **130**, and then the contact module **130** is inserted into the corresponding opening of the tunnel **111**, such that the position limitation can be realized by the barbs and the stop recesses; [0042] b. during the plugging process of the male and female connectors, the contact portion **1411** of the spring **141** can provide elastic support to the contact **131**, such that the contact end **135** of the contact **131** can obtain better elasticity and better contact performance, thereby greatly increasing the plugging and unplugging lifespan of the male and female connectors.

[0043] Through the aforementioned configuration, the contact and the spring form the easy-to-assemble contact module, the first barb of the spring inserted into the terminal hole of the base and coupled with the stop recess realizes the assembly and the limiting of the entire contact module, and thus complex manufacturing processes can be decreased to simplify the assembly process of the product. In addition, the spring can not only be used as a limiting fixture between the contact and the base to ensure that the position of the contact in the base is stable, but also can provide elastic support for the contact, so that the contact can indirectly obtain better elasticity, thereby greatly increasing the plugging and unplugging lifespan of the product.

[0044] The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. An electrical connector, comprising: a base having a plurality of tunnels; and a plurality of contact modules respectively embedded in the tunnels, wherein each of the contact modules comprises a contact and a spring, the spring is located in one of the tunnels and supports a contact end of the contact so that the contact end is suspended, and a back end of the spring has a fixed portion having a first barb, and the first barb hooks into a stop recess of a first tube wall of one of the tunnels.
2. The electrical connector of claim 1, wherein the fixed portion is in a form of an enclosed ring, and the enclosed ring forms a channel to enable the contact to pass through.
3. The electrical connector of claim 2, wherein a surface of the fixed portion in the form of the enclosed ring facing away from the first barb has a second barb, and the second barb hooks into a stop recess of a second tube wall of one of the tunnels.
4. The electrical connector of claim 2, wherein the contact has an ear portion abutting against a sidewall of the fixed portion in the form of the enclosed ring.
5. The electrical connector of claim 4, wherein an opening of the base has a groove corresponding to a shape of the ear portion.
6. The electrical connector of claim 1, wherein a front end of the spring comprises a contact portion suspended from a bottom wall of one of the tunnels, and the spring comprises an extending portion behind the contact portion and at least flat against the bottom wall.
7. The electrical connector of claim 6, wherein each of two sides of the extending portion has a side wing extending in a direction away from the first tube wall.
8. The electrical connector of claim 7, wherein a back of the side wing has a guiding surface.
9. The electrical connector of claim 6, wherein the spring comprises a turning portion connects the

extending portion and corresponding to a step structure of the base.

10. The electrical connector of claim 9, wherein a back of the turning portion connects the fixed portion, and the fixed portion is at least partially flat against the first tube wall.

11. The electrical connector of claim 1, wherein a back of the fixed portion connects a tail portion, and the tail portion extends upward into a positioning recess of a surface of the contact.

12. The electrical connector of claim 1, wherein the spring comprises a contact portion suspended from a bottom wall of one of the tunnels and included by a front end of the spring, an extending portion at least flat against the bottom wall, a turning portion, and a fixed portion in sequence, and the spring is one-piece-formed by a metal plate.

13. The electrical connector of claim 12, wherein the contact end of the contact has a curved structure, and the contact portion of the spring has a curved structure abutting against the curved structure of the contact end of the contact.

14. The electrical connector of claim 12, wherein a surface of the fixed portion facing away from the first barb has a second barb that is aligned with the first barb along a vertical direction.

15. An electrical connector, comprising: a base having a plurality of tunnels; and a plurality of contact modules respectively embedded in the tunnels, wherein each of the contact modules comprises a contact and a spring, the spring is located in one of the tunnels and supports a contact end of the contact so that the contact end is suspended, and a back end of the spring has a fixed portion in a form of an enclosed ring, and the enclosed ring forms a channel to enable the contact to pass through.

16. The electrical connector of claim 15, wherein the contact has an ear portion abutting against a sidewall of the fixed portion in the form of the enclosed ring.

17. The electrical connector of claim 16, wherein an opening of the base has a groove corresponding to a shape of the ear portion.

18. The electrical connector of claim 15, wherein a front end of the spring comprises a contact portion suspended from a bottom wall of one of the tunnels, and the spring comprises an extending portion behind the contact portion and at least flat against the bottom wall.

19. The electrical connector of claim 18, wherein each of two sides of the extending portion has a side wing extending in a direction away from a first tube wall of one of the tunnels.
